

Karan Shihire

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EDUCATION

Modern College of Arts, Science and Commerce

Bachelor of Computer Science (Minor: Embedded Systems & IoT) | CGPA: 8.0

Pune, Maharashtra

Jul 2023 – May 2026

CERTIFICATIONS

Oracle Cloud Infrastructure Foundations Associate

Cloud Architecture, OCI Core Services, Identity & Security

Oracle

PROJECTS

● 1 FaceTrack — Real-Time Attendance / Surveillance System

[GitHub](#)

- Built a real-time face recognition system capable of detecting and identifying multiple faces from live webcam streams using InsightFace and OpenCV.
- Built face enrollment pipeline handling multi-angle capture and embedding extraction, reducing false match rate during testing
 - * Challenge: matching faces reliably across lighting changes → solved using multi-angle enrollment + FAISS cosine similarity threshold tuning
- Developed embedding similarity search workflow using FAISS to perform fast identity matching on high-dimensional facial embeddings.
- Integrated ONNX Runtime inference pipeline to optimize face recognition performance and reduce detection latency during live video processing.
- Developed full-stack AI application integrating computer vision models, backend APIs, database systems, and responsive frontend interfaces.

● 2 NetWatch — Anomaly-Based Intrusion Detection System(IDS)

[GitHub](#)

- Developed an AI-driven intrusion detection system using Isolation Forest for anomaly detection on live network traffic data.
- Built packet capture pipeline using Scapy that monitored TCP/UDP/ICMP traffic and flagged anomalies in under 100ms on local test environment”
- Built a real-time Flask dashboard for visualizing packet statistics, anomaly alerts, and live monitoring data.
- Designed concurrent detection workflow for feature extraction, ML inference, and structured alert generation on real-time traffic streams.

● 3 QuizNet — AI-Powered Quiz Generator

[GitHub](#)

- ”Used Gemini API with structured prompts to generate 3 question types — MCQ, true/false, and reasoning — cutting manual quiz creation time entirely
- Built Django backend with session management, scoring logic, and interactive UI for AI-generated quiz delivery.

TECHNICAL SKILLS

Languages: Python, SQL, Java

AI / ML: Computer Vision, Deep Learning, Scikit-learn, OpenCV, InsightFace, FAISS, ONNX Runtime, NumPy, Pandas

Frameworks / Tools: FastAPI, Pytorch , Flask, React.js, PostgreSQL, Git,

Platforms: GitHub, VS Code, Cursor , Render, Netlify